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FILTER CHANGER

Abstract

A filter changer includes: a base portion; a support shaft portion protruding from the base portion; a turntable supported on the support shaft portion to be rotatable about an axis; a plurality of optical filters arranged on the turntable along a circumferential direction; and a turntable drive portion fixed to the base portion and configured to drive the turntable to rotate around the axis. The turntable drive portion includes: a first drive source part including a first piezoelectric element and a second piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a first abutting part abutting against a part of the turntable and configured to move to depict an elliptical trajectory according to deformation of the first drive source part.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a U.S. National Stage of International Application No. PCT/CN2021/087303, filed on Apr. 14, 2021, the content of which is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a filter changer which is used in a camera module carried in an electronic device (an information device), such as a smart phone, or the like, and with which each of a plurality of optical filters can be arranged on an optical axis of a camera lens.

BACKGROUND

[0003] As a related art, for example, a camera shooting apparatus described in Japanese Laid-open Publication No. 2003-259167 is cited. A filter changer of the camera shooting apparatus includes a stepping motor, a turntable, a plurality of gears, and a worm gear. The stepping motor as a drive source of the filter changer has a screw shaft, and transmits a drive force to the plurality of gears via the worm gear, so as to finally move the turntable with a plurality of optical filters.

SUMMARY

[0004] The present disclosure relates to a filter changer which is used in a camera module carried in an electronic device (an information device), such as a smart phone, or the like, and with which each of a plurality of optical filters can be arranged on an optical axis of a camera lens.

[0005] A first aspect of the present disclosure provides a filter changer, including: a base portion; a support shaft portion protruding from the base portion; a turntable supported on the support shaft portion to be rotatable about an axis; a plurality of optical filters arranged on the turntable along a circumferential direction; and a turntable drive portion fixed to the base portion and configured to drive the turntable to rotate around the axis. The turntable drive portion includes: a first drive source part including a first piezoelectric element and a second piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a first abutting part abutting against a part of the turntable and configured to move to depict an elliptical trajectory according to deformation of the first drive source part.

[0006] A second aspect of the present disclosure provides a camera module including a filter changer. The filter changer includes: a base portion; a support shaft portion protruding from the base portion; a turntable supported on the support shaft portion to be rotatable about an axis; a plurality of optical filters arranged on the turntable along a circumferential direction; and a turntable drive portion fixed to the base portion and configured to drive the turntable to rotate around the axis. The turntable drive portion includes: a first drive source part including a first piezoelectric element and a second piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a first abutting part abutting against a part of the turntable and configured to move to depict an elliptical trajectory according to deformation of the first drive source part.

[0007] A third aspect of the present disclosure provides an electronic device including a filter changer. The filter changer includes: a base portion; a support shaft portion protruding from the base portion; a turntable supported on the support shaft portion to be rotatable about an axis; a plurality of optical filters arranged on the turntable along a circumferential direction; and a turntable drive portion fixed to the base portion and configured to drive the turntable to rotate around the axis. The turntable drive portion includes: a first drive source part including a first piezoelectric element and a second piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a first

abutting part abutting against a part of the turntable and configured to move to depict an elliptical trajectory according to deformation of the first drive source part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a front view schematically showing a structure of a filter changer according to an embodiment of the present disclosure, and perspectively displaying a part thereof.

[0009] FIG. 2 is a longitudinal sectional view schematically showing the structure of the filter changer according to an embodiment of the present disclosure.

[0010] FIG. 3 is a view schematically showing a structure and an action of a drive portion of the filter changer according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0011] A filter changer 1 according to an embodiment of the present disclosure will be described with reference to the drawings. Furthermore, the drawings are schematic views, and relationships between illustrated contents of the drawings may be inaccurate or not matched at times.

[0012] The filter changer 1 according to the embodiment is used in a camera module carried in an electronic device (an information device), such as a smart phone, or the like. As shown in FIGS. 1 and 2, the filter changer 1 according to the embodiment mainly includes a base portion 2, a support shaft portion 3, a turntable 4, a plurality of optical filters 51-54, and a turntable drive portion 6.

[0013] The base portion 2 is a part for supporting each portion constituting the filter changer 1. Although not explicitly shown in the drawings, a space is provided in a part of the base portion 2 consistent with an optical axis of a camera lens. The space is a structure for light passing through the optical filters 51-54 to pass to the camera lens. The optical axis of the camera lens extends in an inward-outward direction of the page of FIG. 1 and an up-down direction of FIG. 2. In the embodiment, an electric wire 9 passes through the base portion 2 to supply power to a filter drive portion 44.

[0014] The support shaft portion 3 is a portion with an axis protruding from the base portion 2 along the optical axis of the camera lens. A lower end of the support shaft portion 3 in the embodiment in FIG. 2 is embedded into and fixed by the base portion 2. In addition, in the embodiment, the electric wire 9 passes through the support shaft portion 3 to supply power to the turntable drive portion 6 and the filter drive portion 44. Furthermore, since the filter drive portion 44 rotates together with the turntable 4, the electrical wire 9 is configured to be conductive while allowing rotation.

[0015] The turntable 4 is supported on the support shaft portion 3 to be rotatable about an axis. The turntable 4 includes a filter support portion 41 and a rotation shaft portion 42, the filter support portion 41 has a circular plate shape and a flat plate shape (i.e., a circular and flat plate shape), and the rotation shaft portion 42 orthogonally extends from a rotation center of the filter support portion 41 to the base portion 2. An outer circumferential surface of the rotation shaft portion 42 has a circular cross-sectional shape with respect to the axis. The rotation shaft portion 42 is mounted on the support shaft portion 3. A ball bearing 7 is arranged between the rotation shaft portion 42 and the support shaft portion 3 and rotates with low friction. The filter support portion 41 rotates integrally with (i.e., together with) the rotation shaft portion 42. Furthermore, the filter support portion 41 is not limited to the circular plate shape, and may have various shapes, such as a polygonal shape, or the like. In addition, the filter support portion 41 and the rotation shaft portion 42 may be formed integrally, or may be formed separately and integrated by assembling.

[0016] As shown in FIG. 1, the plurality of optical filters 51-54 are arranged on the turntable 4 in a circumferential direction. In the embodiment, the four filters are interchangeable. Specifically, the four filters include a polarizing filter (a CPL filter) 51, a dimming filter (a ND filter) 52, a diffusion

filter (a filter for generating a blurred image) **53**, and a hole portion **54** (no filter). Furthermore, in the embodiment, the hole portion **54** is also described as a type of filter. The four filters **51-54** are arranged at a certain angle interval (specifically, 90°) in the circumferential direction. By rotating the turntable **4**, the filters **51-54** are respectively positioned on the optical axis of the camera lens every 90° rotation angle. Specifically, a state in which the polarizing filter (the CPL filter) **51** is located on the optical axis of the camera lens is used as a reference. As shown in FIG. **1**, when the turntable **4** is rotated counterclockwise by 90° , the dimming filter (the ND filter) **52** is located on the optical axis of the camera lens. When the turntable **4** is rotated counterclockwise by 90° again, the diffusion filter **53** is located on the optical axis of the camera lens. When the turntable **4** is rotated counterclockwise by 90° again, the hole portion **54** is located on the optical axis of the camera lens (in a state without the optical filter).

[0017] Rotation of the turntable **4** is detected by a combination of a Hall element **81** (refer to FIG. **2**, but only one is shown in FIG. **2**) arranged on the base portion **2** and a permanent magnet **82** (refer to FIG. **1**) arranged on the turntable **4**. Thus, a positional relationship of each of the filters **51-54** with respect to the camera lens can be detected. In the embodiment, the rotation of the turntable **4** can be easily detected by the combination of the Hall element **81** and the permanent magnet **82** of an existing unit. Furthermore, in contrast to the embodiment, the permanent magnet **82** may be arranged on the base portion **2**, and the Hall element **81** may be arranged on the turntable **4**. The Hall element **81** is supplied with power by a flexible circuit board not shown in the drawings. Two Hall elements **81**, which are separated by 90° with the rotation center of the filter support portion **41** of the turntable **4** as a reference, serve as a group (refer to FIG. **2** for the Hall element **81**), and two permanent magnets **82**, which are separated by 90° with the rotation center of the filter support portion **41** of the turntable **4** as a reference, serve as a group (refer to FIG. **1** for the permanent magnet **82**). As shown in table 1, it can be detected which filter is located on the optical axis of the camera lens by a mode in which either of the two Hall elements **81** (A, B) reacts to the permanent magnet. In table 1, a state in which the Hall element **81** reacts is “ON”, and a state in which the Hall element **81** does not react is “OFF”. Furthermore, table 1 only shows an example, and the present disclosure is not limited thereto.

TABLE-US-00001 TABLE 1 Position Hall A Hall B CPL ON ON Hole OFF ON Bokeh OFF OFF ND ON OFF

[0018] The turntable drive portion **6** is fixed on the base portion **2** and drives the turntable **4** (specifically, the rotation shaft portion **42**) to rotate around the axis. In the embodiment, as shown in FIG. **1**, two turntable drive portions **6** are arranged to have a substantial V shape in a front view. Furthermore, the number of the turntable drive portions **6** is not limited to two, and may be one, three or more. FIG. **3** schematically shows a structure and an action of the turntable drive portion **6**. The turntable drive portion **6** includes a first drive source part **61** and a first abutting part **62**. Three protrusions arranged on an illustrated upper end edge of the turntable drive portion **6** are electrodes **63** for supplying power to piezoelectric elements **611**, **612**. The piezoelectric element is supplied with power by a flexible circuit board not shown in the drawings. At least one of the turntable drive portions **6** includes a support spring **64** at a radially outer position, and a force is applied in a radially inward direction by the support spring **64**.

[0019] The first drive source part **61** is formed by attaching a first piezoelectric element **611** and a second piezoelectric element **612**, which are configured to be elongated by energization, in a direction orthogonal to their elongation directions. In some embodiments, the first piezoelectric element **611** and the second piezoelectric element **612** may have the same elongation direction, that is, the elongation directions of the first piezoelectric element **611** and the second piezoelectric element **612** may be identical. The first piezoelectric element **611** and the second piezoelectric element **612** are also arranged in a diagonal direction, and two first piezoelectric elements **611** and two second piezoelectric elements **612** are used to form the drive source portion **61**. In the embodiment, since the drive source portion **61** may be bent and deformed by the energization of

the first piezoelectric element **611** and the second piezoelectric element **612**, either the first piezoelectric element **611** or the second piezoelectric element **612** is energized only, and the first piezoelectric element **611** and the second piezoelectric element **612** are not energized at the same time. However, a voltage may be set so that both the first piezoelectric element **611** and the second piezoelectric element **612** can be energized at the same time.

[0020] The first abutting part **62** is a protrusion-shaped part which is located at a left end shown in FIG. **3** and protrudes in a length direction of the first drive source part **61**. The first abutting part **62** is arranged at a boundary between the first piezoelectric element **611** and the second piezoelectric element **612** on the left side in the drawing (FIG. **3**). The first abutting part **62** is a part which outputs the deformation of the first drive source part **61** as a force. Therefore, a positional relationship of the first abutting part **62** with respect to the first drive source part **61** is not limited to the embodiment, and the first abutting part **62** may also be arranged on a side surface of the first drive source part **61** or biased with respect to the first piezoelectric element **611** and the second piezoelectric element **612**. The first abutting part **62** abuts against the outer circumferential surface of the rotation shaft portion **42** which is a part of the turntable **4**, and is configured to move to depict elliptical trajectories **V1**, **V2** shown by dotted lines in FIG. **3** according to the deformation of the first drive source part **61**.

[0021] When power is supplied to the first piezoelectric elements **611** located in shaded upper left and lower right regions in FIG. **3**, the regions extend in a horizontal direction. On the other hand, when no power is supplied to the second piezoelectric elements **612** located in colorless lower left and upper right regions in FIG. **3**, the regions maintain rectangular shapes shown in the drawing. Therefore, the first piezoelectric element **611** is deformed in a direction depicted by arrows in the drawing, and the second piezoelectric element **612** is also deformed accordingly. Therefore, a left portion of an assembly of the piezoelectric elements (an assembly of the two first piezoelectric elements **611** and the two second piezoelectric elements **612** in the embodiment) of the drive source portion **61** in the drawing is bent and deformed into an inverse U shape, and a right portion thereof in the drawing is bent and deformed into a U shape, so that the entire assembly is deformed into a “~” shape (or an inverted and horizontal S shape). Therefore, the first abutting part **62** abutting against the turntable **4** moves downwards and leftwards in FIG. **3**. Since this movement is carried out while the first piezoelectric element **611** extends, this movement is a movement of an upper left half of the elliptical trajectory **V1** shown by the lower left dotted line in FIG. **3**. The turntable **4** pressed by the first abutting part **62** is rotated clockwise (rightwards) in FIG. **1** under an action of a force related to this movement. In this way, in the embodiment, the turntable **4** may be directly driven by the turntable drive portion **6**.

[0022] Next, when no power is supplied to the first piezoelectric elements **611** located in the shaded upper left and lower right regions, the regions are contracted in the horizontal direction. The assembly of the piezoelectric elements in the first drive source part **61** returns from the “~” shape to the original rectangular shape in the drawing. At this point, the first abutting part **62** moves upwards and rightwards in FIG. **3**. Since this movement is carried out while the first piezoelectric element **611** is contracted, this movement is a movement of a lower right half of the elliptical trajectory **V1** shown by the dotted line in FIG. **3**. As described above, by turning on and off power supply to the first piezoelectric element **611**, the first abutting part **62** depicts the elliptical trajectory shown in the lower left part of FIG. **3**.

[0023] Conversely, when power is supplied to the second piezoelectric elements **612** located in the colorless lower left and upper right regions, the assembly of the piezoelectric elements in the first drive source part **61** is deformed into a horizontal S shape contrary to the above shape, and therefore, the first abutting part **62** depicts the elliptical trajectory **V2** shown in the upper left part of FIG. **3**. The turntable **4** subjected to a force related to this movement rotates counterclockwise (leftwards) in FIG. **1**. In this way, the turntable **4** can be rotated clockwise and counterclockwise according to the energization states of the piezoelectric elements **611**, **612** arranged in the turntable

drive portion **6**. Furthermore, a central portion **613** of the assembly of the four piezoelectric elements **611**, **612** does not move even when each of the piezoelectric elements **611**, **612** is energized. Therefore, for example, when the first drive source part **61** is fixed outside, it may be fixed at the central portion **613** as a stationary point.

[0024] A movable filter which is a part of the plurality of optical filters is arranged to be rotatable with respect to the turntable **4**, and has a circular outer circumference shape. In the embodiment, the movable filter is the polarizing filter (the CPL filter) **51**. Since the polarizing filter (the CPL filter) **51** may have a varying polarizing effect depending on a position in the circumferential direction with respect to the optical axis of the camera lens, an image captured by a camera can be changed. Since the polarizing filter (the CPL filter) **51** is rotated with respect to the turntable **4**, a rotation support portion **43** and the filter drive portion **44** are provided at a part of the turntable **4** where the polarizing filter (the CPL filter) **51** is arranged. An annular bracket **511** is provided at an outer edge of the polarizing filter (the CPL filter) **51**. In order to rotate a filter body part (a part through which light passes) with respect to the turntable **4** by contact with the rotation support portion **43**, the bracket **511** rotates around the filter body part. The rotation support portion **43** includes a plurality of spheres in the embodiment, and rotatably supports the bracket **511**. The rotation support portion **43** in the embodiment is formed by two spheres. The filter drive portion **44** rotates the polarizing filter (the CPL filter) **51**.

[0025] The filter drive portion **44** has a same structure and a same action principle as the turntable drive portion **6**, and includes: a second drive source part **441** formed by attaching a first piezoelectric element and a second piezoelectric element (not shown, but the same as the piezoelectric elements shown in FIG. 3), which are configured to be elongated by energization, in a direction orthogonal to their elongation directions; and a second abutting part **442** abutting against an outer circumferential edge of the bracket **511** and configured to move to depict an elliptical trajectory according to deformation of the second drive source part **441**. According to the energization states of the piezoelectric elements in the filter drive portion **44**, the polarizing filter (the CPL filter) **51** can be rotated clockwise and counterclockwise. In this way, in the embodiment, the polarizing filter (the CPL filter) **51** can be directly driven by the filter drive portion **44**.

[0026] As described above, the embodiment provides the filter changer **1**, and the filter changer **1** includes: the base portion **2**; the support shaft portion **3** protruding from the base portion **2**; the turntable **4** supported on the support shaft portion **3** to be rotatable about the axis; the plurality of optical filters **51-54** arranged on the turntable **4** along the circumferential direction; and the turntable drive portion **6** fixed to the base portion **2** and configured to drive the turntable **4** to rotate around the axis. The turntable drive portion **6** includes: the first drive source part **61** formed by attaching the first piezoelectric element **611** and the second piezoelectric element **612**, which are configured to be elongated by energization, in the direction orthogonal to their elongation directions; and the first abutting part **62** abutting against the part of the turntable **4** and configured to move to depict the elliptical trajectories **V1**, **V2** according to the deformation of the first drive source part **61**.

[0027] With this structure, the turntable **4** can be directly driven by the turntable drive portion **6** which generates the drive force by the piezoelectric elements **611**, **612**.

[0028] In addition, the movable filter **51** which is the part of the plurality of optical filters **51-54** is arranged to be rotatable with respect to the turntable **4**, and has the circular outer circumference shape, and the turntable **4** has the filter drive portion **44** for rotating the movable filter **51**. The filter drive portion **44** may include: the second drive source part **441** formed by attaching the first piezoelectric element and the second piezoelectric element, which are configured to be elongated by energization, in the direction orthogonal to their elongation directions; and the second abutting part **442** abutting against the outer circumferential edge of the movable filter **51** and configured to move to depict the elliptical trajectory according to the deformation of the second drive source part **441**.

[0029] With this structure, the movable filter **51** can be directly driven by the filter drive portion **44** which generates the drive force by the piezoelectric elements.

[0030] In addition, the movable filter **51** may be the polarizing filter (the CPL filter) **51**.

[0031] With this structure, the polarizing filter (the CPL filter) **51** can be easily rotated by using the filter drive portion, and the polarizing filter has different polarizing effects according to the position in the circumferential direction with respect to the optical axis of the camera lens.

[0032] In addition, the rotation of the turntable **4** can be detected by the Hall element **81** arranged on one of the base portion **2** and the turntable **4** and the permanent magnet **82** arranged on the other of the base portion **2** and the turntable **4**.

[0033] With this structure, the rotation of the turntable **4** can be detected easily by an existing method.

[0034] In addition, the support shaft portion **3** is the portion with the axis protruding from the base portion along the optical axis of the camera lens, the electric wire **9** for supplying power to the turntable drive portion **6** passes through the support shaft portion, and the electric wire **9** can ensure to be conductive even when the turntable **4** rotates.

[0035] In addition, the turntable **4** includes the flat-plate-shaped filter support portion **41** and the rotation shaft portion **42**, the rotation shaft portion **42** orthogonally extends from the rotation center of the filter support portion **41** to the base portion **2**, and the rotation shaft portion **42** may be mounted to the support shaft portion **3** by the bearing **7**.

[0036] In addition, the plurality of optical filters **51-54** may be arranged on the turntable **4** at a certain angle interval in the circumferential direction.

[0037] In addition, the first abutting part **62** may be the protrusion-shaped part which is located at the end of the turntable drive portion **6** and protrudes in the length direction of the first drive source part **61**, and may be arranged at the boundary of the first piezoelectric element **611** and the second piezoelectric element **612**.

[0038] In addition, the present disclosure provides the camera module using the filter changer **1**.

[0039] In addition, the present disclosure provides the electronic device using the filter changer **1**.

[0040] As described above, the filter changer **1** according to the embodiment may be formed in a simple structure directly driven by the piezoelectric elements **611**, **612**. Therefore, a low cost and miniaturization can be realized. In addition, by using the high-output piezoelectric element, an intermediate transmission unit, such as a gear, or the like, and a speed reduction unit are not required, and the high-speed rotation can be carried out by direct driving. Therefore, the desired optical filters **51-54** can be quickly positioned on the optical axis of the camera lens, and thus the filter changer **1** with a high performance can be provided.

Claims

1. A filter changer, comprising: a base portion; a support shaft portion protruding from the base portion; a turntable supported on the support shaft portion to be rotatable about an axis; a plurality of optical filters arranged on the turntable along a circumferential direction; and a turntable drive portion fixed to the base portion and configured to drive the turntable to rotate around the axis, wherein the turntable drive portion comprises: a first drive source part comprising a first piezoelectric element and a second piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a first abutting part abutting against a part of the turntable and configured to move to depict an elliptical trajectory according to deformation of the first drive source part.
2. The filter changer according to claim 1, wherein the plurality of optical filters comprises a movable filter, and the movable filter is arranged to be rotatable with respect to the turntable, and has a circular outer circumference shape; the turntable has a filter drive portion for rotating the movable filter; and the filter drive portion comprises: a second drive source part comprising a third

- piezoelectric element and a fourth piezoelectric element configured to be elongated by energization, in a direction orthogonal to their elongation directions; and a second abutting part abutting against an outer circumferential edge of the movable filter and configured to move to depict an elliptical trajectory according to deformation of the second drive source part.
- 3.** The filter changer according to claim 2, wherein the movable filter is a polarizing filter.
- 4.** The filter changer according to claim 1, wherein a permanent magnet is arranged on one of the base portion and the turntable and a Hall element is arranged on the other of the base portion and the turntable to detect rotation of the turntable.
- 5.** The filter changer according to claim 1, wherein the support shaft portion is a portion with an axis protruding from the base portion along an optical axis of a camera lens, and an electric wire for supplying power to the turntable drive portion passes through the support shaft portion; and the electric wire is configured to be conductive in case that the turntable rotates.
- 6.** The filter changer according to claim 1, wherein the turntable comprises a flat-plate-shaped filter support portion and a rotation shaft portion, and the rotation shaft portion orthogonally extends from a rotation center of the filter support portion to the base portion; and the rotation shaft portion is mounted to the support shaft portion by a bearing.
- 7.** The filter changer according to claim 1, wherein the plurality of optical filters are arranged on the turntable at a certain angle interval in the circumferential direction.
- 8.** The filter changer according to claim 2, wherein the first abutting part is a protrusion-shaped part located at an end of the turntable drive portion and protruding in a length direction of the first drive source part, and is arranged at a boundary of the first piezoelectric element and the second piezoelectric element.
- 9.** A camera module, comprising a filter changer, wherein the filter changer comprises: a base portion; a support shaft portion protruding from the base portion; a turntable supported on the support shaft portion to be rotatable about an axis; a plurality of optical filters arranged on the turntable along a circumferential direction; and a turntable drive portion fixed to the base portion and configured to drive the turntable to rotate around the axis, wherein the turntable drive portion comprises: a first drive source part comprising a first piezoelectric element and a second piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a first abutting part abutting against a part of the turntable and configured to move to depict an elliptical trajectory according to deformation of the first drive source part.
- 10.** An electronic device, comprising a filter changer, wherein the filter changer comprises: a base portion; a support shaft portion protruding from the base portion; a turntable supported on the support shaft portion to be rotatable about an axis; a plurality of optical filters arranged on the turntable along a circumferential direction; and a turntable drive portion fixed to the base portion and configured to drive the turntable to rotate around the axis, wherein the turntable drive portion comprises: a first drive source part comprising a first piezoelectric element and a second piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a first abutting part abutting against a part of the turntable and configured to move to depict an elliptical trajectory according to deformation of the first drive source part.
- 11.** The electronic device according to claim 10, wherein the plurality of optical filters comprises a movable filter, and the movable filter is arranged to be rotatable with respect to the turntable, and has a circular outer circumference shape; the turntable has a filter drive portion for rotating the movable filter; and the filter drive portion comprises: a second drive source part comprising a third piezoelectric element and a fourth piezoelectric element configured to be elongated by energization, and attached in a direction orthogonal to their elongation directions; and a second abutting part abutting against an outer circumferential edge of the movable filter and configured to move to depict an elliptical trajectory according to deformation of the second drive source part.

- 12.** The electronic device according to claim 11, wherein the movable filter is a polarizing filter.
- 13.** The electronic device according to claim 10, wherein a permanent magnet is arranged on one of the base portion and the turntable and a Hall element is arranged on the other of the base portion and the turntable to detect rotation of the turntable.
- 14.** The electronic device according to claim 10, wherein the support shaft portion is a portion with an axis protruding from the base portion along an optical axis of a camera lens, and an electric wire for supplying power to the turntable drive portion passes through the support shaft portion; and the electric wire is configured to be conductive in case that the turntable rotates.
- 15.** The electronic device according to claim 10, wherein the turntable comprises a flat-plate-shaped filter support portion and a rotation shaft portion, and the rotation shaft portion orthogonally extends from a rotation center of the filter support portion to the base portion; and the rotation shaft portion is mounted to the support shaft portion by a bearing.
- 16.** The electronic device according to claim 10, wherein the plurality of optical filters are arranged on the turntable at a certain angle interval in the circumferential direction.
- 17.** The electronic device according to claim 11, wherein the first abutting part is a protrusion-shaped part located at an end of the turntable drive portion and protruding in a length direction of the first drive source part, and is arranged at a boundary of the first piezoelectric element and the second piezoelectric element.
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